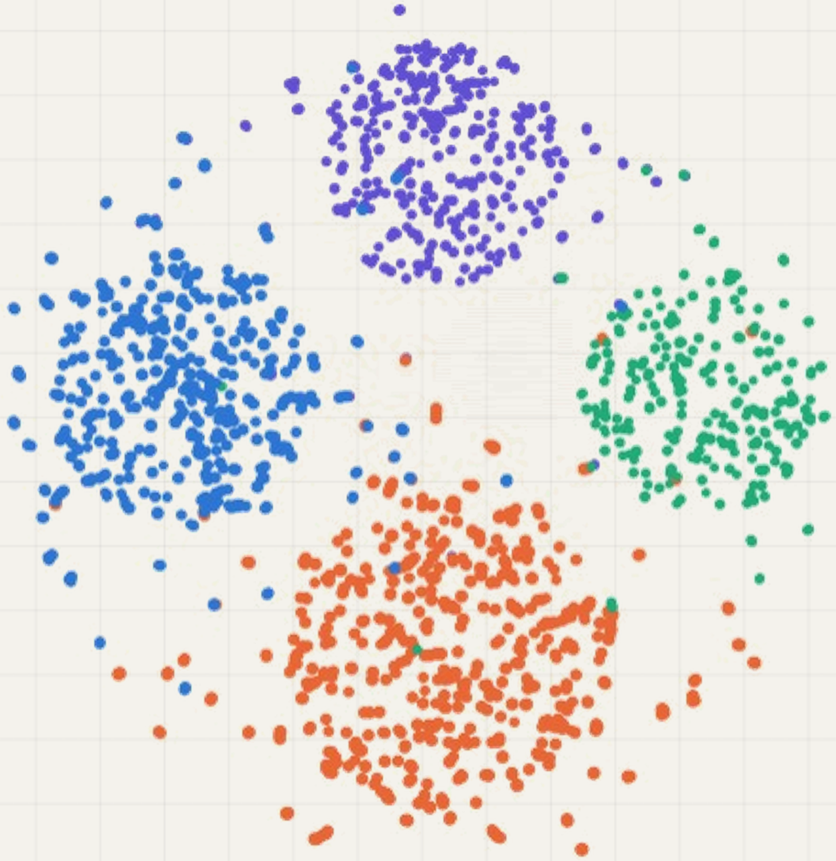
TEAM DUMBOCRACY

Clout.

An Election Game

NAME	ROLL NUMBER
Pakki Sai Kaushik	AI23BTECH11017
G Madhava Asrith Murthy	AI23BTECH11007
Vuppula Hemanth Reddy	AI23BTECH11033
Vishank Singh	AI23BTECH11032
B U Deekshith Patel	AI23BTECH11003



MOTIVATION

Campaigns spend big to move opinions they can't see

1 **Opinions drift** DEGROOT CONSENSUS

People slowly come to think like their friends.

2 **Influence spreads** INDEPENDENT CASCADE

A message passed on can die out or take off.

3 **Campaigns work blind** EPISTEMIC LOGIC

A party never knows exactly what voters want.

INDIA, 2019 ELECTION

₹60,000 cr

spent on one general election, mostly to change what people think.

Centre for Media Studies

THE GAME

Win a majority in five rounds, without seeing the voters

01

Setup

A hidden electorate of 10,000 voters is drawn. You get 500 coins.

02

Survey

Buy questions, then commit a **manifesto**: fixed for the game.

03 · ×5 rounds

Campaign

Hire party workers, air media. Voters drift towards their friends.

repeat each round

04

Election

More than half the votes wins. Then the hidden truth is revealed.

10 coins
Survey question

20 coins
Party worker

~50 coins
Media airing

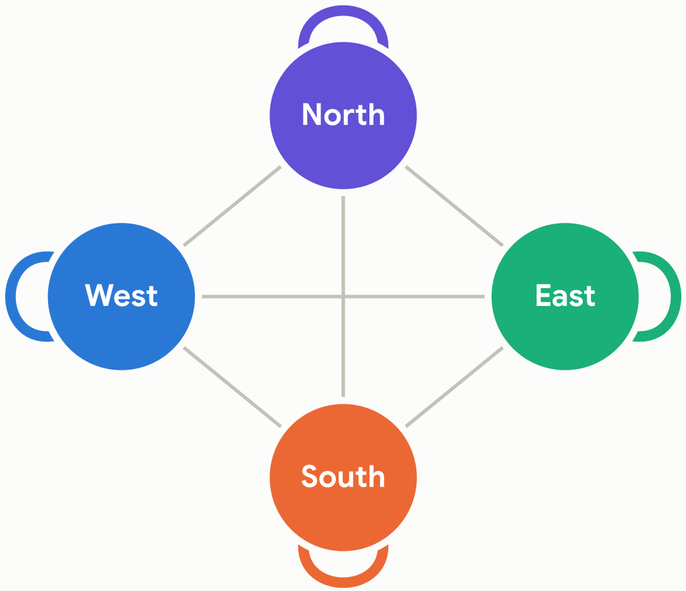
MANIFESTO = A BUDGET SPLIT



Education · Welfare · Agriculture · Industry · Environment

THE ELECTORATE

Four regions, one friendship network, hidden tastes



Thick loops: ~¾ of friendships stay inside a region.

WANTS

A budget split

Each region has a hidden archetype, e.g. *agrarian* or *industrial*.

MOVED BY

Friends, media or canvassing

Each region has a hidden personality.

ZEALOTS

5% never change

But their friends still listen to them.

VOTES

Only when close enough

Leans your way once it agrees with **82.5%** of your budget.

THE ELECTORATE · IN MATHS

How we build a world

REGIONS	$n_r = m + \big[\mathrm{Mult}(N - 4m,\ \pi)\big]_r, \quad m = \lceil 0.15N \rceil$	Each region gets **≥ 15%**; the rest split at random
FRIENDSHIPS	$P_{rr} = s\,p_r, \ \ P_{rs} = s\,p_{rs}, \quad p_r \sim U[0.5, 0.9],\ p_{rs} \sim U[0.01, 0.2]$	Stochastic block model; **s** sets mean degree 30
WANTS	$v_i(0) \sim \mathrm{Dir}\big(\kappa\,\mu_{\sigma(r)}\big), \quad \kappa = 60, \ \ v_i \in \Delta^5$	Budget split around the region's hidden **archetype σ**
MOVED BY	$\alpha_i,\ \beta_i,\ \gamma_i \sim \mathrm{Beta}\big(\tau(r)\big), \quad \Pr[\text{zealot}] = 0.05$	Friends, media, canvassing via hidden **personality τ**
DRIFT	$v_i \leftarrow (1 - a_i - c_i)\,v_i + a_i\,\bar{v}_{\mathcal{N}(i)} + c_i\,b, \quad a_i = 0.15\,\alpha_i$	Damped **DeGroot** step; **c** is the campaign pull
VOTES	$P_i = \mathrm{sig}\big(20\,(S_i - 0.825)\big), \quad S_i = 1 - \tfrac{1}{2}\lVert v_i - b \rVert_1$	Agreement with manifesto **b** sets the vote chance

HIDDEN TYPE $\theta = (\sigma, \tau)$: $360 \times 24 = 8640$ worlds — What the player must infer from surveys and polls.

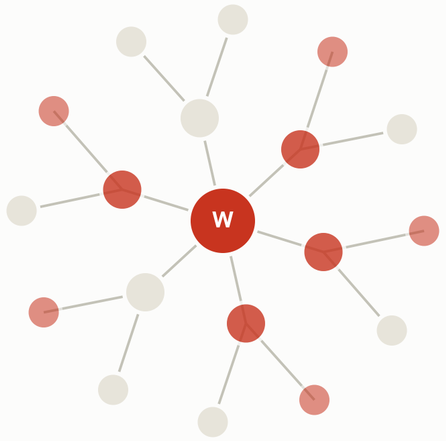
TWO WAYS TO SPEND

Workers start cascades. Media gives a nudge.

Workers `20 coins`

- A real voter, pinned to your manifesto
- Converts friends, **one hop per round**
- Hire early: more rounds to spread

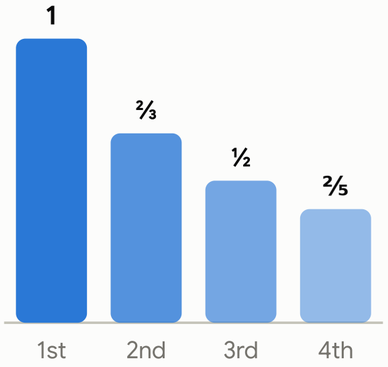
Takes off when $\lambda_{max}(A \cdot D_q) > 1$



Media `~50 coins`

- Reaches **60%** of a region at once
- A temporary pull, no network needed
- Each repeat airing is weaker

Strength ∝ `1 / (1 + 0.5·n)`



airing in the same region

LINK The manifesto matters for both: voters who already like it convert more easily.

EPISTEMIC & TEMPORAL LOGIC

Knowing a win exists ≠ knowing how to win

The player can't see the world, so we track every world that **matches what they have seen**.

K true in **every** such world: the player *knows* it

M true in **some** world: it is *possible*

After each game, the debrief shows these verdicts for every round, alongside the hidden truth.

Knows *that*

every world has some winning plan

	π1	π2	π3
w1	1	0	0
w2	0	1	0
w3	0	0	1

No single plan is safe

Knows *how*

one plan wins in every world

	π1	π2	π3
w1	1	1	0
w2	0	1	0
w3	0	1	1

Play π2: guaranteed victory

Guaranteed ⇒ Knows a win exists ⇒ Possible neither arrow reverses

RESULTS · 100 WORLDS EACH

What wins an election in Clout

MANIFESTO ALONE

0.37

Best possible vote share on day one. **No manifesto wins without a campaign.**

ONE WORKER

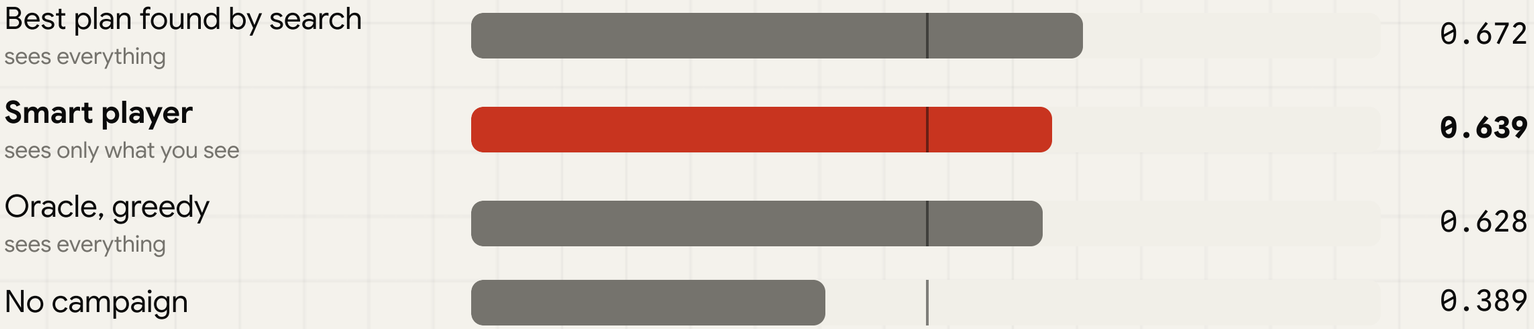
62%

of the maximum gain. Five workers give 93%: **buy a few, early.**

WHEN VS WHERE

~10×

Timing (round 1 vs 5) beats placement: **cascades cross regions.**



PLANNING BEATS SEEING

The smart player sees **less** than the Oracle but wins **98%** of games by looking ahead.

Line marks a 0.5 vote share.

TAKEAWAYS

Three lessons from 23 experiments

1 Aim for election day

Drift alone adds ~0.11 to a manifesto built for the final voters.

2 A few workers, then media

Workers saturate fast; early media is the better buy after that.

3 Planning beats seeing

Looking ahead matters more than seeing the hidden world.

UP NEXT

Live demo.

Setup → Survey → Campaign → Debrief

Next.js client · FastAPI engine · PyTorch simulator

LIMITATIONS

What our results do not tell you

Synthetic voters	Tuned so the game is playable, not fitted to a real electorate.
One party	No opponent, the manifesto is fixed, friendships never change.
Heuristic baselines	The Oracle and the plan search are strong, but not proven optimal.
Sampled knowledge	Logic verdicts come from a finite sample of smaller worlds.
No human data	Every result comes from computer players; the playtest was dropped.

TEAM DUMBOCRACY

Individual contributions

G. Madhava Asrith Murthy AI23BTECH11007	Built the smart player (surveys, manifesto, planning) and ran the strategy experiments.
B. U. Deekshith Patel AI23BTECH11003	Built the electorate: regions, friendship network, preferences, zealots, drift and voting.
Vuppula Hemanth Reddy AI23BTECH11033	Designed what the player sees and buys: polls, surveys, workers, media and their prices.
Pakki Sai Kaushik AI23BTECH11017	Built the epistemic and temporal logic, the game engine and the GPU experiment harness.
Vishank Singh AI23BTECH11032	Built the web game: its screens, visualisations, debrief timeline and web API.

TEAM DUMBOCRACY

Thank you.

Questions?

github.com/saikaushhikp/CLOUT

Pakki Sai Kaushik · G. Madhava Asrith Murthy · Vuppula Hemanth Reddy · Vishank Singh · B. U. Deekshith Patel